



EMERSON UNIVERSITY MULTAN

Submitted To: Mam Iqra Iqbal Khan
Group No # 1

Group Members

Name	Roll No #
Ali Muzamil	05
Muhammad Khubaib (GL)	24
Muhammad Danish	11
Muhammad Ammar	31
Muhammad Usman	43

**Topic: E-Commerce Product and Order
Management System**

Semantic Objects of E-Commerce Product and Order Management System

Below is the explanation of all 12 objects

CUSTOMER

SELLER

PRODUCT

CATAGORY

ORDER

PAYMENT

CART

CART-ITEM

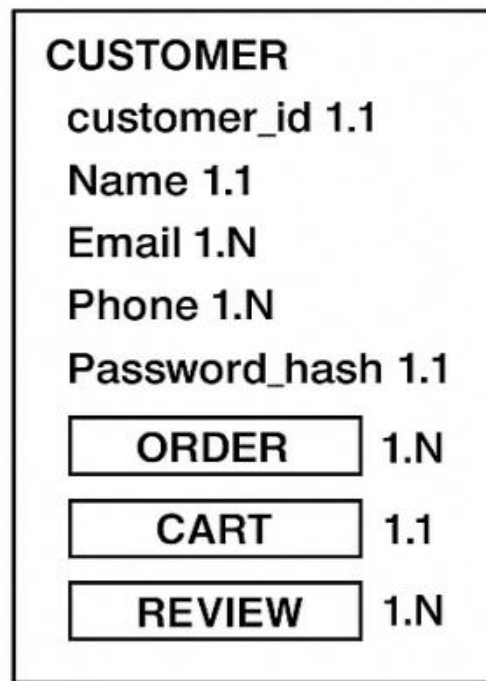
ORDER-ITEM

INVENTORY

SHIPMENT

REVIEW

Semantic Objects with Attributes and Relationships: -



The Customer is the person who uses the application to create an account, place orders, give reviews, save their address, and add items to the cart. A customer is a user who buys products. Their info like name, email, and password is stored. They can place orders, have a cart, and write reviews.

SELLER

seller_id 1.1

seller_name 1.1

Email 1.N

business_name 1.1

A seller provides products on the platform. Their business and contact info are stored. A seller can list Many products.

PRODUCT

product_id 1.1

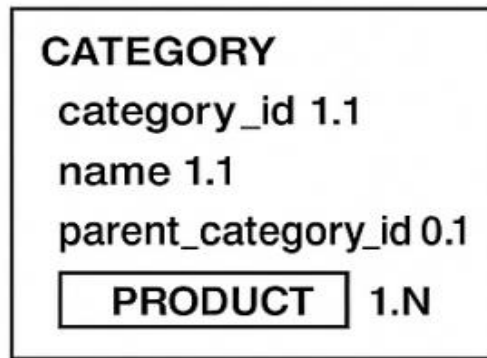
Name 1.1

Price 1.N

category_id 1.N

seller_id 1.1

A product is an item available for purchase. It has a price, name, and belongs to a category and seller



A category of groups with similar products. It can also have a parent category.
Every product belongs to one category.



A cart holds items a customer wants to buy. Each customer has one cart,
containing many cart items.

ORDER

order_id 1.1

customer_id 1.1

order_date 1.1

total_amount 1.N

payment_id 1.1

An order is created when a customer purchases items. It includes date, total amount, and payment info.

ORDER

order_itemid 1.1

order_id 1.1

product_id 11

payment_1.N

payment_date 1.

An order is created when a customer purchases items. It includes date, total amount, and payment info.

PAYMENT

payment_id 1.1

order_id 1.1

amount 1.N

payment_mehod 1.N

payment_status 1.1

Payment stores how a customer pays for an order. It includes the amount, status, method, and date.

INVENTORY

inventory_id 1.1

product_id 1.1

stock_quantity 1.N

restock_date 1.N

warehouse_location

Inventory tracks are available in stock for each product. Stores quantity, restock date, and warehouse Location.

SHIPMENT

shipment_id 1

order_id 1.1

carrier_name 1.1

shipment_status 1.N

delivery_date 1.N

The shipment details how an order will be delivered. Includes carrier, status, and delivery date. Delivery contains the tracking and timing details of how the order is delivered

REVIEW

review_id 1.1

customer_id 1.1

product_id 1.1

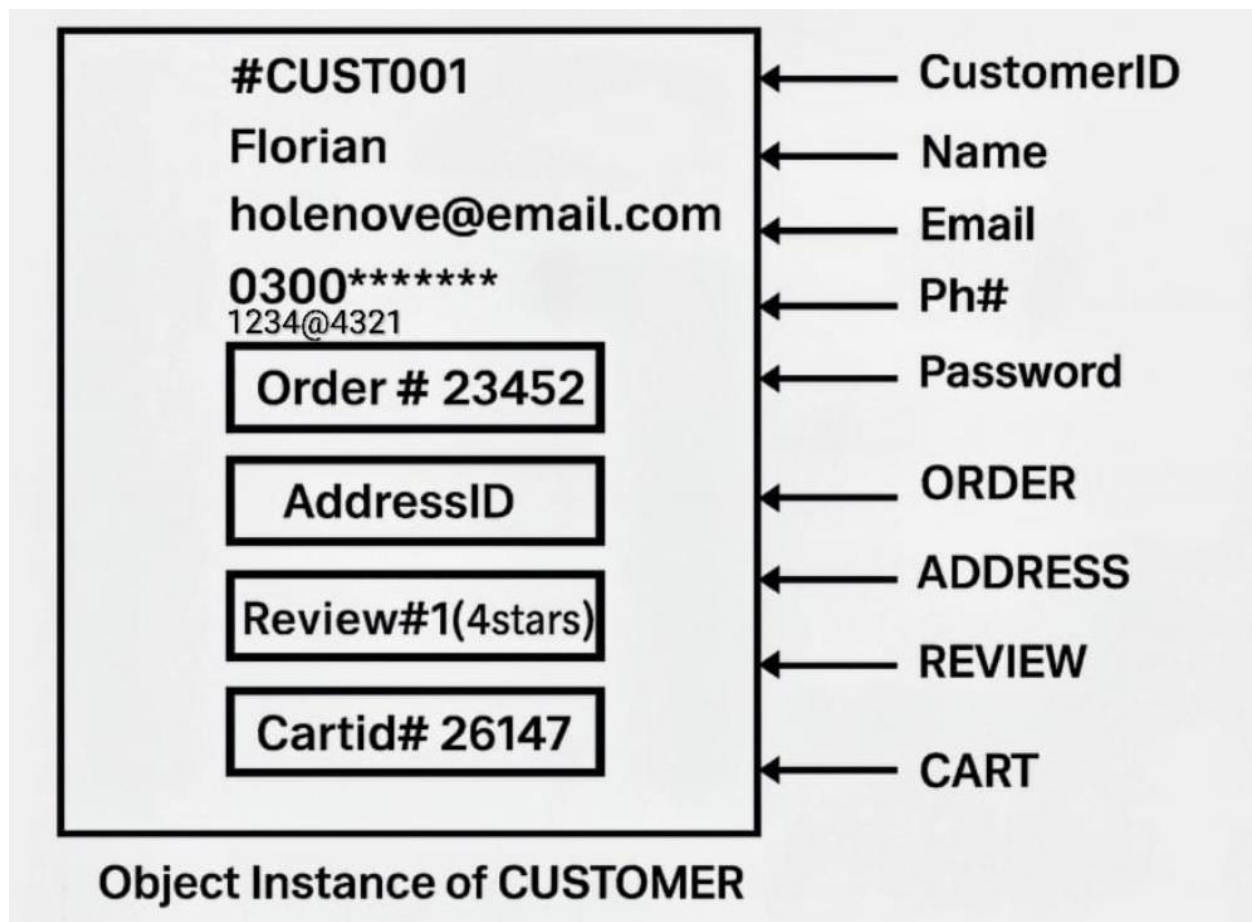
rating 1.N

review_text 1.N

review_date 1.N

A review is feedback written by a customer about a product. It includes rating, text, and date

Semantic object instances: -



This represents **one specific CUSTOMER object** created from the *Customer* class. Every attribute inside it belongs to that single customer. These show links to other objects connected to this customer



Object Instance of ADDRESS

This represents one specific ADDRESS object created from the *Address* class. The ADDRESS object stores the customer's address separately.

Semantic Object View (Example) :-

Customer Shipment View:-

Sr	CustomerID	Name	Ph #	Order#	AddressID
1	#CUST001	Florian	0300*****	23452	(Associated Address)

The **Object Instance** for a customer named Florian, which contains many attributes, such as personal details, order information, and passwords.

Customer Address View:-

SR#	Street	City	Postal Code	Customer ID
1	Garden Town	Multan	60000	#25699

ADDRESS object you provided and integrate it into a new, single view of the **CUSTOMER** object, specifically focusing on a scenario like a **Billing Statement** view.

